

BMAT-001: Mathematics I (Common for All Branches)

Department of Mathematics • Maharishi Markandeshwar (Deemed to be University), Mullana (Ambala)

UNIT I (25%)Matrices, Rank, Normal Form, Gauss-Jordan Inverse, $AX=B$ Consistency & Cayley-Hamilton.**UNIT II (25%)**

Rolle's & LMVT, Taylor/Maclaurin Expansions, Curvature, L'Hospital & Beta-Gamma Functions.

UNIT III (25%)

Multivariable Calculus, Euler's Theorem, Extrema, Lagrange Multipliers, Gradient, Divergence & Curl.

UNIT IV (25%)

Infinite Series Convergence Tests (Ratio, Raabe, Log, Root, Leibnitz) & Fourier Series Expansions.

UNIT I: MATRICES, RANK, SYSTEM OF EQUATIONS & EIGENVALUE THEOREMSRank, Row-Echelon Form, Normal Form, Gauss-Jordan Inversion, $AX = B$ Consistency & Cayley-Hamilton Theorem**1. Rank of a Matrix, Row Echelon & Normal Form (PAQ = $[I_r \ 0 \ 0]$)****Rank of a Matrix:** The rank $\rho(A)$ is the maximum number of linearly independent row or column vectors in A . A matrix has rank r if it contains at least one non-zero minor of order r , and all minors of order $(r+1)$ and higher vanish identically.**Normal Form (Canonical Form):** Every $m \times n$ matrix of rank r can be reduced to one of the canonical block forms: $[I_r]$, $[I_r \ 0]$, $[I_r; 0]$, or $[I_r \ 0; 0 \ 0]$ using both elementary row and column transformations. There exist non-singular square matrices P ($m \times m$) and Q ($n \times n$) such that: $P A Q = [I_r \ 0 \ 0]$.**Gauss-Jordan Matrix Inversion Algorithm:**

- Form augmented matrix $[A \mid I_n]$.
- Apply elementary row operations to reduce left block A to the Identity matrix I_n .
- The transformed right block is identically the matrix inverse: $[A \mid I_n] \xrightarrow{\text{Row Ops}} [I_n \mid A^{-1}]$.

2. Consistency of Linear Systems ($AX = B$) & Eigenvalue Theorems**Rouché-Capelli Theorem:** Form augmented matrix $[A \mid B]$.

- Inconsistent (No Solution):** $\rho(A) \neq \rho(A|B)$.
- Consistent (Unique Solution):** $\rho(A) = \rho(A|B) = n$ (Number of unknowns).
- Consistent (Infinite Solutions):** $\rho(A) = \rho(A|B) = r < n$. The solution space possesses $(n - r)$ linearly independent free parameters.

Eigenvalues & Cayley-Hamilton Theorem:

- Characteristic Equation:** $|A - \lambda I| = 0 \Leftrightarrow \lambda^3 - S_1 \lambda^2 + S_2 \lambda - |A| = 0$.
- Cayley-Hamilton Theorem:** Every square matrix satisfies its own characteristic equation: $A^3 - S_1 A^2 + S_2 A - |A| I = 0$.
- Inverse Computation:** Post-multiplying by A^{-1} yields: $A^{-1} = \frac{1}{|A|} [A^2 - S_1 A + S_2 I]$.

UNIT II: DIFFERENTIAL & INTEGRAL CALCULUS, CURVATURE & BETA-GAMMA FUNCTIONS

Rolle's, LMVT, Taylor's Series, Curvature Formulae, L'Hospital & Euler's Integrals

1. Mean Value Theorems & Power Series Expansions

Rolle's Theorem: If $f(x)$ is (1) continuous in $[a, b]$, (2) differentiable in (a, b) , and (3) $f(a) = f(b)$, then there exists at least one point $c \in (a, b)$ such that the tangent is horizontal: $f'(c) = 0$.

Lagrange's Mean Value Theorem (LMVT): If $f(x)$ satisfies (1) and (2), then there exists $c \in (a, b)$ such that: $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Taylor's & Maclaurin's Expansions:

• Taylor's Series about $x = a$: $f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \dots$

• Maclaurin's Series (about $x = 0$):

$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$ | $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$

$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$ | $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$

2. Curvature, Radius of Curvature (ρ) & Beta-Gamma Functions

Curve Representation	Exact Radius of Curvature Formula: ρ	Key Analytical Condition
Cartesian $y = f(x)$	$\rho = \frac{[1 + (y_1')^2]^{3/2}}{y_2'}$	Where $y_1' = \frac{dy}{dx}$, $y_2' = \frac{d^2y}{dx^2} \neq 0$.
Parametric $x=x(t)$, $y=y(t)$	$\rho = \frac{[x_1'^2 + y_1'^2]^{3/2}}{ x_1' y_2' - y_1' x_2' }$	Primes denote derivatives w.r.t. parameter t .
Polar $r = f(\theta)$	$\rho = \frac{[r^2 + r_1'^2]^{3/2}}{r^2 + 2 r_1' r_2' - r r_2'}$	Where $r_1' = \frac{dr}{d\theta}$, $r_2' = \frac{d^2r}{d\theta^2}$.

Beta & Gamma Functions (Euler's Integrals):

- $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx$ ($n > 0$) • $\Gamma(n+1) = n \Gamma(n) = n!$ • $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.
- $B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx = 2 \int_0^{\pi/2} \sin^{2m-1}\theta \cos^{2n-1}\theta d\theta$.
- Fundamental Connection:** $B(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$.
- Duplication Formula:** $\Gamma(m) \Gamma(\frac{m+1}{2}) = \frac{\sqrt{\pi}}{2^{2m-1}} \Gamma(2m)$.

UNIT III: MULTIVARIABLE CALCULUS, EXTREMA & VECTOR DIFFERENTIAL CALCULUS

Euler's Homogeneous Theorem, Two-Variable Extrema, Lagrange Multipliers, Gradient, Divergence & Curl

1. Euler's Theorem on Homogeneous Functions & Two-Variable Extrema

Euler's Theorem: If $u(x, y)$ is a homogeneous function of degree n in x and y , then:

1. $x (\partial u / \partial x) + y (\partial u / \partial y) = n u.$
2. $x^2 (\partial^2 u / \partial x^2) + 2 x y (\partial^2 u / \partial x \partial y) + y^2 (\partial^2 u / \partial y^2) = n (n - 1) u.$

Maxima and Minima of Functions of Two Variables:

Let $f_x = 0, f_y = 0$ determine stationary points (a, b) . Compute $r = f_{xx}, s = f_{xy}, t = f_{yy}$ and discriminant $\Delta = rt - s^2$:

- If $rt - s^2 > 0$ and $r < 0$: Local Maximum at (a, b) .
- If $rt - s^2 > 0$ and $r > 0$: Local Minimum at (a, b) .
- If $rt - s^2 < 0$: **Saddle Point** (Stationary point with no extreme value).
- If $rt - s^2 = 0$: Inconclusive; higher-order terms must be investigated.

2. Vector Differential Calculus: Gradient, Directional Derivative, Divergence & Curl

Operator / Quantity	Mathematical Coordinate Expression	Physical Meaning & Null Tests
Gradient $\nabla \phi$	$(\partial \phi / \partial x) \hat{i} + (\partial \phi / \partial y) \hat{j} + (\partial \phi / \partial z) \hat{k}$	Vector normal to level surface $\phi = c$; points in direction of maximum rate of increase.
Directional Derivative	$D_a \phi = \nabla \phi \cdot \hat{a} = \nabla \phi \cdot \frac{\mathbf{a}}{ \mathbf{a} }$	Rate of change of scalar field ϕ in direction of unit vector $\hat{a}$.
Divergence $\nabla \cdot \mathbf{F}$	$(\partial F_1 / \partial x) + (\partial F_2 / \partial y) + (\partial F_3 / \partial z)$	Net volumetric outflow flux per unit volume. $\nabla \cdot \mathbf{F} = 0 \Leftrightarrow$ Solenoidal Vector Field .
Curl $\nabla \times \mathbf{F}$	$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$	Rotational circulation density (vorticity). $\nabla \times \mathbf{F} = 0 \Leftrightarrow$ Irrotational / Conservative Field ($\mathbf{F} = \nabla \phi$).

UNIT IV: CONVERGENCE OF SEQUENCES & SERIES AND FOURIER SERIES

Ratio, Raabe's, Logarithmic, Cauchy Root, Alternating Leibnitz Rule & Fourier Expansion Theorems

1. Infinite Series Convergence Tests

Convergence Test	Limit Ratio Expression: L	Convergence Criterion
D'Alembert's Ratio Test	$L = \lim_{n \rightarrow \infty} (u_n / u_{n+1})$	$L > 1 \Rightarrow$ Convergent; $L < 1 \Rightarrow$ Divergent; $L = 1 \Rightarrow$ Fails.
Raabe's Test (Higher Ratio)	$L = \lim_{n \rightarrow \infty} n [(u_n / u_{n+1}) - 1]$	$L > 1 \Rightarrow$ Convergent; $L < 1 \Rightarrow$ Divergent; $L = 1 \Rightarrow$ Fails.
Logarithmic Test	$L = \lim_{n \rightarrow \infty} [n \ln((u_n / u_{n+1}))]$	$L > 1 \Rightarrow$ Convergent; $L < 1 \Rightarrow$ Divergent.
Cauchy's Root Test	$L = \lim_{n \rightarrow \infty} (u_n)^{1/n}$	$L < 1 \Rightarrow$ Convergent; $L > 1 \Rightarrow$ Divergent; $L = 1 \Rightarrow$ Fails.
Leibnitz's Test (Alternating)	$\sum_{n=1}^{\infty} (-1)^{n-1} u_n$	Convergent if: (1) $u_{n+1} \leq u_n$ for all n, and (2) $\lim_{n \rightarrow \infty} u_n = 0$.

2. Fourier Series, Dirichlet Conditions & Half-Range Expansions

Dirichlet's Conditions: A periodic function f(x) of period 2L can be expanded in a convergent Fourier series if in [-L, L]: (1) f(x) is single-valued and bounded, (2) f(x) has a finite number of points of discontinuity, (3) f(x) has a finite number of maxima and minima.

Euler's Formulae in Interval [-L, L]:
 $f(x) = (a_0 / 2) + \sum_{n=1}^{\infty} [a_n \cos(\frac{n\pi x}{L}) + b_n \sin(\frac{n\pi x}{L})]$
• $a_0 = \frac{1}{L} \int_{-L}^L f(x) dx$ • $a_n = \frac{1}{L} \int_{-L}^L f(x) \cos(\frac{n\pi x}{L}) dx$ • $b_n = \frac{1}{L} \int_{-L}^L f(x) \sin(\frac{n\pi x}{L}) dx$.
Even Function [$f(-x) = f(x)$]: $b_n = 0 \Rightarrow$ Fourier Cosine Series: $a_0 = \frac{2}{L} \int_0^L f dx$, $a_n = \frac{2}{L} \int_0^L f \cos \frac{n\pi x}{L} dx$.
Odd Function [$f(-x) = -f(x)$]: $a_0 = 0$, $a_n = 0 \Rightarrow$ Fourier Sine Series: $b_n = \frac{2}{L} \int_0^L f \sin \frac{n\pi x}{L} dx$.

MMDU END-SEMESTER EXAMINATION PAPER (SOLVED WITH DUAL DIAGRAMS)

Batch 2025–2029 • BMAT-001: Mathematics I • Total Marks: 60 • 100% Dual Diagrams

SECTION A: 12 COMPULSORY CONCEPTUAL QUESTIONS (1 MARK EACH = 12 MARKS)

Question 1 (Unit 1): Rank Invariance under Elementary Operations

1 Mark

State why elementary row transformations do not alter the rank of a matrix.

Question Transformation Vector: Matrix A [Row Vectors R_1, R_2] $\rightarrow$ [Row Operation: $R_2 \leftarrow R_2 - k R_1$] $\rightarrow$ Span(Rows) preserved.

STEP-BY-STEP SOLUTION:

Solution Row Space Isomorphism:

$\text{Span}\{R_1, R_2, \dots, R_m\} = \text{Span}\{R_1, R_2 - k R_1, \dots, R_m\} \Rightarrow \text{Dimension of Row Space} = \rho(A)$ (Invariant).

Elementary transformations correspond to multiplying by non-singular elementary matrices (which have non-zero determinant). Because multiplying by invertible matrices preserves vector space dimension, the rank remains strictly invariant.

Question 2 (Unit 1): Eigenvalues of a Nilpotent Matrix

1 Mark

Prove that every eigenvalue of a nilpotent matrix ($A^k = 0$ for some $k \geq 1$) is identically zero.

Question Nilpotent Operator: $A^k X = 0$ where $AX = \lambda X$ and $X \neq 0$.

STEP-BY-STEP SOLUTION:

Solution Spectral Power Derivation:

$AX = \lambda X \Rightarrow A^k X = \lambda^k X = 0 \Rightarrow \lambda^k = 0 \Rightarrow \lambda = 0$.

Because the eigenvector X is non-zero, the scalar λ^k must equal zero, proving that all eigenvalues of any nilpotent matrix are strictly zero.

Question 3 (Unit 1): Cayley-Hamilton Inverse Criterion

1 Mark

Under what condition can the Cayley-Hamilton theorem be used to evaluate the inverse matrix A^{-1} ?

Question Determinant Axis: Non-Singular $|A| \neq 0$ [Invertible] vs Singular $|A| = 0$ [Non-Invertible].

STEP-BY-STEP SOLUTION:

Solution Constant Term Extraction:

$A^n + c_{n-1} A^{n-1} + \dots + (-1)^n |A| I = 0 \Rightarrow A^{-1} = ((-1)^{n-1} / |A|) [A^{n-1} + c_{n-1} A^{n-2} + \dots]$ (Demands $|A| \neq 0$).

A must be non-singular (determinant $|A| \neq 0$). If $|A| = 0$, the constant term vanishes and division by zero is undefined, preventing inverse extraction.

Question 4 (Unit 2): Geometric Interpretation of Rolle's Theorem

1 Mark

State the geometric meaning of the point c in Rolle's Theorem.

Question Tangent Geometry: Curve $y = f(x)$ between $(a, f(a))$ and $(b, f(b))$ where $f(a) = f(b) \rightarrow$ Horizontal Tangent at c .

STEP-BY-STEP SOLUTION:

Solution Tangent Slope Zero Vector:

Slope $m = f'(c) = 0 \Rightarrow$ Tangent line at $(c, f(c))$ is strictly parallel to the x -axis.

Geometric meaning: If a smooth continuous curve crosses the same horizontal level at two points, there must exist at least one intermediate point where the curve turns and its tangent is parallel to the x -axis.

Question 5 (Unit 2): Radius of Curvature of a Straight Line

1 Mark

Compute the curvature κ and radius of curvature ρ for the straight line $y = mx + c$.

Question Straight Line Slope: $y_1 = m$ (Constant) $\rightarrow y_2 = \frac{d^2y}{dx^2} = 0$.

STEP-BY-STEP SOLUTION:

Solution Curvature Formula Substitution:

$\rho = \left(\frac{1 + m^2}{0} \right) = \infty \Rightarrow \kappa = \frac{1}{\rho} = 0$.

A straight line does not bend; its curvature is identically zero, and its radius of curvature is infinite (∞).

Question 6 (Unit 2): Value of Gamma $\frac{1}{2}$ **1 Mark**

Evaluate $\Gamma \frac{1}{2}$ using the Gaussian probability integral.

Question Integral Substitution: $\Gamma \frac{1}{2} = \int_{-\infty}^{\infty} e^{-x} x^{-1/2} dx$. Substitute $x = u^2$.

STEP-BY-STEP SOLUTION:

Solution Gaussian Integral Formulation:

$$\Gamma \frac{1}{2} = 2 \int_0^{\infty} e^{-u^2} du = 2 \times \frac{\sqrt{\pi}}{2} = \sqrt{\pi} \approx 1.772.$$

Via transformation to the Gaussian integral, $\Gamma \frac{1}{2} = \sqrt{\pi}$.

Question 7 (Unit 3): Homogeneous Degree via Euler's Theorem**1 Mark**

Find the degree of homogeneity for $u(x, y) = \frac{x^3 + y^3}{x - y}$ and evaluate $x u_x + y u_y$.

Question Scaling Test: Replace x with tx , y with $ty \Rightarrow u(tx, ty) = t^n u(x, y)$.

STEP-BY-STEP SOLUTION:

Solution Euler Scaling Derivation:

$$u(tx, ty) = (t^3(x^3 + y^3) / t(x - y)) = t^2 u(x, y) \Rightarrow \text{Degree } n = 2 \Rightarrow x u_x + y u_y = 2 u.$$

The function is homogeneous of degree $n = 3 - 1 = 2$. By Euler's theorem, $x (\partial u / \partial x) + y (\partial u / \partial y) = 2 u$.

Question 8 (Unit 3): Physical Meaning of Solenoidal Vector Fields**1 Mark**

State the physical condition for a fluid velocity field V to be solenoidal.

Question Flux Balance Volume: Inflow Flux = Outflow Flux across any closed bounding surface S .

STEP-BY-STEP SOLUTION:

Solution Incompressible Continuity Equation:

$$\nabla \cdot V = 0 \Rightarrow \text{Zero net volumetric divergence} \Rightarrow \text{Incompressible Fluid Flow.}$$

A solenoidal vector field has zero divergence ($\nabla \cdot V = 0$), physically representing an incompressible fluid with no sources or sinks.

Question 9 (Unit 3): Conservative Vector Field and Scalar Potential**1 Mark**

What condition guarantees that a force field F is conservative (work done is path-independent)?

Question Closed Loop Work: $\oint F \cdot dr = 0$ along any closed contour C .

STEP-BY-STEP SOLUTION:

Solution Irrotational Curl Criterion:

$$\nabla \times F = 0 \Rightarrow F = \nabla \phi \Rightarrow \int_A^B F \cdot dr = \phi(B) - \phi(A).$$

The force field must be irrotational ($\nabla \times F = 0$), which guarantees the existence of a scalar potential ϕ such that work done depends solely on endpoints.

Question 10 (Unit 4): Failure of Ratio Test on p-Series**1 Mark**

Why does D'Alembert's Ratio Test fail to determine the convergence of the p-series $\sum (1 / n^p)$?

Question Limit Ratio: $u_n / u_{n+1} = \frac{1 + 1}{n}^p$ as $n \rightarrow \infty$.

STEP-BY-STEP SOLUTION:

Solution Indeterminate Limit Ratio:

$$L = \lim_{n \rightarrow \infty} (u_n / u_{n+1}) = \lim_{n \rightarrow \infty} (1 + \frac{1}{n})^p = 1^p = 1 \Rightarrow \text{Test Fails for ALL } p!$$

Because the limit ratio evaluates identically to 1 for every real exponent p , the ratio test provides no diagnostic information (Raabe's or integral test must be applied).

Question 11 (Unit 4): Fourier Coefficients for an Odd Function**1 Mark**

State the values of Fourier coefficients a_0 and a_n for an odd function $f(x)$ in $[-\pi, \pi]$.

Question Parity Symmetry: $f(-x) = -f(x)$ across origin in symmetrical interval $[-\pi, \pi]$.

STEP-BY-STEP SOLUTION:

Solution Symmetric Integral Cancellation:

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = 0 \text{ (Odd integrand)}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx = 0 \text{ (Odd} \times \text{Even} = \text{Odd integrand)}$$

For any odd function, $a_0 = 0$ and $a_n = 0$ identically. The Fourier expansion contains purely sine terms (Fourier Sine Series).

Question 12 (Unit 4): Alternating Harmonic Series Convergence**1 Mark**

Classify the series $\sum (-1)^{n-1} / n$ as conditionally or absolutely convergent.

Question Convergence Split: $\sum (-1)^{n-1}/n$ converges to $\ln(2)$ | $\sum |(-1)^{n-1}/n| = \sum \frac{1}{n}$ diverges.

STEP-BY-STEP SOLUTION:

Solution Dual Test Classification:

1. Leibnitz Test: $u_n = 1/n \Rightarrow u_{n+1} \leq u_n$ and $\lim \frac{1}{n} = 0 \Rightarrow$ Convergent!

2. Absolute Series: $\sum |u_n| = \sum \frac{1}{n} \Rightarrow$ Divergent harmonic series ($p=1$)

Because the alternating series converges but its absolute series diverges, it is strictly **Conditionally Convergent**.

SECTION B: 4 ANALYTICAL PROBLEMS (2 MARKS EACH = 8 MARKS)

Question 13 (Unit 1): Rank and Normal Form of a 3x3 Matrix

2 Marks

Reduce matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$ to normal form and determine its rank.

Question Matrix Structure: Rows $R_2 = 2 R_1$ and $R_3 = 3 R_1$ are scalar multiples of R_1 .

STEP-BY-STEP SOLUTION:

Solution Row & Column Reduction Trace:

$$R_2 \rightarrow R_2 - 2 R_1 \Rightarrow [0, 0, 0]$$

$$R_3 \rightarrow R_3 - 3 R_1 \Rightarrow [0, 0, 0] \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$C_2 \rightarrow C_2 - 2 C_1, C_3 \rightarrow C_3 - 3 C_1 \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} I_1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The normal form is $[I_1, 0; 0, 0]$, proving the matrix has **Rank $\rho(A) = 1$** .

Question 14 (Unit 2): Radius of Curvature of Catenary Curve

2 Marks

Find the radius of curvature ρ for the catenary $y = c \cosh(x/c)$.

Question Catenary Hanging Geometry: Hyperbolic curve $y = c \cosh(x/c)$ with derivatives $y_1 = \sinh(x/c)$, $y_2 = \frac{1}{c} \cosh(x/c)$.

STEP-BY-STEP SOLUTION:

Solution Hyperbolic Identity Simplification:

$$1 + (y_1)^2 = 1 + \sinh^2(x/c) = \cosh^2(x/c)$$

$$\rho = \left(\frac{[1 + y_1^2]^{3/2}}{y_2} \right) = \left(\frac{[\cosh^2(x/c)]^{3/2}}{\frac{1}{c} \cosh(x/c)} \right) = \left(\frac{\cosh^3(x/c)}{\frac{1}{c} \cosh(x/c)} \right) = c \cosh^2(x/c) = \frac{y^2}{c}$$

The radius of curvature simplifies elegantly to $\rho = \frac{y^2}{c}$.

Question 15 (Unit 3): Directional Derivative along Unit Vector

2 Marks

Find the directional derivative of $\phi = x^2 y z + 4 x z^2$ at point $(1, -2, -1)$ in the direction of vector $2i - j - 2k$.

Question Normal & Direction Vectors: $\nabla \phi$ at $(1, -2, -1)$ projected onto unit vector $a / |a|$.

STEP-BY-STEP SOLUTION:

Solution Vector Projection Formulation:

$$\nabla \phi = (2xyz + 4z^2)i + (x^2 z)j + (x^2 y + 8xz)k$$

$$\text{At } (1, -2, -1): \nabla \phi = (4 + 4)i + (-1)j + (-2 - 8)k = 8i - j - 10k$$

$$\text{Unit vector: } \hat{a} = (2i - j - 2k) / \sqrt{4 + 1 + 4} = \frac{2i - j - 2k}{3}$$

$$D_a \phi = (8i - j - 10k) \cdot \frac{2i - j - 2k}{3} = \frac{16 + 1 + 20}{3} = \frac{37}{3}$$

The directional derivative evaluates to **37 / 3**.

Question 16 (Unit 4): Cauchy's Root Test on Infinite Series

2 Marks

Test the convergence of the series $\sum \left[\left(\frac{n}{n+1} \right)^n \times \frac{1}{2^n} \right]$ using Cauchy's Root Test.

Question nth Power Term: General term u_n has overall exponent n .

STEP-BY-STEP SOLUTION:

Solution nth Root Limit Trace:

$$(u_n)^{1/n} = \left[\left(\frac{n}{n+1} \right)^n \cdot \frac{1}{2^n} \right]^{1/n} = \frac{n}{n+1} \cdot \frac{1}{2} = \left(1 / 1 + 1/n \right) \cdot \frac{1}{2}$$

$$L = \lim_{n \rightarrow \infty} (u_n)^{1/n} = \frac{1}{1 + 0} \cdot \frac{1}{2} = \frac{1}{2} < 1$$

Because limit ratio $L = \frac{1}{2} < 1$, the series is strictly **Convergent**.

SECTION C: 4 COMPREHENSIVE PROBLEMS (4 MARKS EACH = 16 MARKS)

Question 17 (Unit 1): Complete Consistency Analysis of Linear System ($AX = B$)

4 Marks

Investigate the consistency of the system: $x + y + z = 6$, $x + 2y + 3z = 10$, $x + 2y + \lambda z = \mu$. Determine values of λ and μ for: (1) No solution, (2) Unique solution, (3) Infinite solutions.

Question Augmented System Matrix: $[A | B]$ with parameters λ and μ in third row.

STEP-BY-STEP SOLUTION:

Solution Echelon Reduction:

$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{array} \right] \xrightarrow{[R_2-R_1, R_3-R_1]} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda-1 & \mu-6 \end{array} \right] \xrightarrow{[R_3-R_2]} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda-3 & \mu-10 \end{array} \right]$

1. No Solution: $\lambda = 3$ and $\mu \neq 10 \Rightarrow \rho(A) = 2 \neq \rho(A|B) = 3$
2. Unique Solution: $\lambda \neq 3$ (any μ) $\Rightarrow \rho(A) = \rho(A|B) = 3 = n$
3. Infinite Solutions: $\lambda = 3$ and $\mu = 10 \Rightarrow \rho(A) = \rho(A|B) = 2 < n$ (3)

The system has: No solution when $\lambda=3, \mu \neq 10$; Unique solution when $\lambda \neq 3$; Infinite solutions when $\lambda=3, \mu=10$.

Question 18 (Unit 2): Radius of Curvature for Parametric Cycloid

4 Marks

For the cycloid $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$, prove that the radius of curvature is $\rho = 4a \cos \frac{\theta}{2}$.

Question Cycloid Parametric Derivatives: $x' = a(1 + \cos\theta)$, $y' = a \sin\theta \rightarrow x'' = -a \sin\theta$, $y'' = a \cos\theta$.

STEP-BY-STEP SOLUTION:

Solution Trigonometric Parametric Trace:

$$x'^2 + y'^2 = a^2(1 + \cos\theta)^2 + a^2 \sin^2\theta = a^2(1 + 2\cos\theta + \cos^2\theta + \sin^2\theta) = 2a^2(1 + \cos\theta) = 4a^2 \cos^2 \frac{\theta}{2}$$

$$x' y'' - y' x'' = a(1 + \cos\theta)(a \cos\theta) - (a \sin\theta)(-a \sin\theta) = a^2(\cos\theta + \cos^2\theta + \sin^2\theta) = a^2(1 + \cos\theta) = 2a^2 \cos^2 \frac{\theta}{2}$$

$$\rho = \left(\frac{[x'^2 + y'^2]^{3/2}}{|x' y'' - y' x''|} \right) = \left(\frac{[4a^2 \cos^2 \frac{\theta}{2}]^{3/2}}{2a^2 \cos^2 \frac{\theta}{2}} \right) = \left(\frac{8a^3 \cos^3 \frac{\theta}{2}}{2a^2 \cos^2 \frac{\theta}{2}} \right) = 4a \cos \left(\frac{\theta}{2} \right)$$

The radius of curvature is verified identically as $\rho = 4a \cos \frac{\theta}{2}$.

Question 19 (Unit 3): Two-Variable Extrema Determination

4 Marks

Find and classify all local extrema and saddle points of the function: $f(x, y) = x^3 + y^3 - 3axy$ (where $a > 0$).

Question Stationary System: $f_x = 3x^2 - 3ay = 0$ and $f_y = 3y^2 - 3ax = 0$.

STEP-BY-STEP SOLUTION:

Solution Hessian Discriminant Classification:

From $f_x = 0 \Rightarrow y = x^2 / a$. Substitute in $f_y = 0 \Rightarrow x^4 / a^2 - ax = 0 \Rightarrow x(x^3 - a^3) = 0$

Critical Points: $(0, 0)$ and (a, a)

Second Derivatives: $r = f_{xx} = 6x$, $s = f_{xy} = -3a$, $t = f_{yy} = 6y$

At $(0, 0)$: $r = 0$, $s = -3a$, $t = 0 \Rightarrow \Delta = rt - s^2 = 0 - 9a^2 = -9a^2 < 0 \Rightarrow$ SADDLE POINT!

At (a, a) : $r = 6a > 0$, $s = -3a$, $t = 6a \Rightarrow \Delta = (6a)(6a) - (-3a)^2 = 36a^2 - 9a^2 = 27a^2 > 0$

Since $\Delta > 0$ and $r > 0 \Rightarrow$ LOCAL MINIMUM at (a, a) with value $f(a, a) = -a^3$

Point $(0, 0)$ is a saddle point; point (a, a) is a local minimum with minimum value $-a^3$.

Question 20 (Unit 4): Raabe's Test on Higher-Order Series

4 Marks

Test the convergence of the series: $1 + \frac{1}{2} \cdot \frac{1}{3} + \frac{1 \cdot 3}{2 \cdot 4} \cdot \frac{1}{5} + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \cdot \frac{1}{7} + \dots$

Question General Term Expression: $u_n = \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{2 \cdot 4 \cdot 6 \dots (2n)} \times [1 / (2n+1)]$.

STEP-BY-STEP SOLUTION:

Solution Ratio Failure & Raabe's Resolution:

$$(u_n / u_{n+1}) = \frac{2n+2}{2n+1} \cdot \frac{2n+3}{2n+1} = \frac{4n^2 + 10n + 6}{4n^2 + 4n + 1} \Rightarrow \lim_{n \rightarrow \infty} (u_n / u_{n+1}) = 1 \text{ (Ratio Test FAILS!)}$$

$$\text{Apply Raabe's Test: } L = \lim_{n \rightarrow \infty} n \left[(u_n / u_{n+1}) - 1 \right] = \lim_{n \rightarrow \infty} n \left[\frac{4n^2 + 10n + 6}{4n^2 + 4n + 1} - 1 \right] = \lim_{n \rightarrow \infty} \frac{6n^2 + 5n}{4n^2 + 4n + 1} = \frac{6}{4} = \frac{3}{2}$$

Since $L = \frac{3}{2} = 1.5 > 1 \Rightarrow$ CONVERGENT by Raabe's Test!

Because the Raabe limit $L = 1.5 > 1$, the infinite series converges.

SECTION D: 4 EXTENDED ESSAY & DERIVATION PROBLEMS (6 MARKS EACH = 24 MARKS)

Question 21 (Unit 1): Verification of Cayley-Hamilton Theorem & Inverse / Higher Power

6 Marks

For matrix $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 3 & -1 & 1 \end{bmatrix}$. (1) Verify Cayley-Hamilton Theorem, (2) Evaluate A^{-1} , (3) Compute A^4 .

Question Characteristic Equation: $|A - \lambda I| = -\lambda^3 + S_1 \lambda^2 - S_2 \lambda + |A| = 0$.

STEP-BY-STEP SOLUTION:

Solution Cayley-Hamilton Full Matrix Derivation:

1. Invariants: $S_1 = \text{Tr}(A) = 1 + 1 + 1 = 3$

$S_2 = M_{11} + M_{22} + M_{33} = (1 - 1) + (1 - 3) + (1 - 0) = 0 - 2 + 1 = -1$

$|A| = 1(0) - 2(3) + 1(-3) = -9$

Char Eq: $\lambda^3 - 3\lambda^2 - \lambda + 9 = 0 \Rightarrow A^3 - 3A^2 - A + 9I = 0$

2. Evaluate $A^2 = \begin{bmatrix} 4 & 3 & 0 \\ -3 & 2 & -2 \\ 6 & 4 & 5 \end{bmatrix}$, $A^3 = \begin{bmatrix} 4 & 11 & 1 \\ -9 & -2 & -7 \\ 21 & 11 & 7 \end{bmatrix}$

Check: $A^3 - 3A^2 - A + 9I = 0 \Rightarrow \text{VERIFIED!}$

3. Inverse: $A^{-1} = -\frac{1}{9}[A^2 - 3A - I] = \frac{1}{9}\begin{bmatrix} 0 & 3 & 3 \\ 3 & 2 & -1 \\ 3 & -7 & -1 \end{bmatrix}$

4. Higher Power: $A^4 = 3A^3 + A^2 - 9A$

Cayley-Hamilton is verified with zero remainder matrix, yielding the exact inverse and fourth power matrices.

Question 22 (Unit 2): Derivation of Relation between Beta and Gamma Functions

6 Marks

State and prove the fundamental transformation relation connecting Beta and Gamma functions: $B(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$. Hence evaluate $\int_0^{\pi/2} \sqrt{\tan \theta} d\theta$.

Question Double Integral Transformation: $\Gamma(m)\Gamma(n) = 4 \iint e^{-(x^2+y^2)} x^{2m-1} y^{2n-1} dx dy$ in polar coordinates.

STEP-BY-STEP SOLUTION:

Solution Polar Coordinate Substitution Trace:

$x = r \cos \theta$, $y = r \sin \theta$, $dx dy = r dr d\theta$

$\Gamma(m)\Gamma(n) = 4 \int_0^{\pi/2} \cos^{2m-1} \theta \sin^{2n-1} \theta d\theta \times \int_0^\infty e^{-r^2} r^{2(m+n)-1} dr = B(m, n) \times \Gamma(m+n)$

$\Rightarrow B(m, n) = (\Gamma(m)\Gamma(n) / \Gamma(m+n))$ (PROVED!)

Application: $\int_0^{\pi/2} \sin^{1/2} \theta \cos^{1/2} \theta d\theta = \frac{1}{2} B\left(\frac{3}{4}, \frac{1}{4}\right) = \frac{1}{2} \left(\Gamma\left(\frac{3}{4}\right)\Gamma\left(\frac{1}{4}\right) / \Gamma(1)\right)$

Reflection Formula: $\Gamma(p)\Gamma(1-p) = (\pi / \sin(p\pi)) \Rightarrow \Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{3}{4}\right) = (\pi / \sin\left(\frac{\pi}{4}\right)) = \pi\sqrt{2}$

Result = $\frac{1}{2} \times \pi\sqrt{2} = \frac{\pi}{\sqrt{2}}$

The fundamental Beta-Gamma connection is proven, evaluating the integral to $\frac{\pi}{\sqrt{2}}$.

Question 23 (Unit 3): Constrained Optimization via Lagrange Multipliers

6 Marks

Find the maximum and minimum distances from the origin to the surface of the ellipsoid: $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ (where $a > b > c > 0$) using the Method of Lagrange Multipliers.

Question Objective & Constraint: $f(x, y, z) = x^2 + y^2 + z^2$ subject to $\phi(x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 = 0$.

STEP-BY-STEP SOLUTION:

Solution Lagrangian Gradient System:

$F = (x^2 + y^2 + z^2) + \lambda \left[\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 \right]$

$(\partial F / \partial x) = 2x \frac{1+\lambda}{a^2} = 0 \Rightarrow x = 0 \text{ or } \lambda = -a^2$

$(\partial F / \partial y) = 2y \frac{1+\lambda}{b^2} = 0 \Rightarrow y = 0 \text{ or } \lambda = -b^2$

$(\partial F / \partial z) = 2z \frac{1+\lambda}{c^2} = 0 \Rightarrow z = 0 \text{ or } \lambda = -c^2$

Case 1: $\lambda = -a^2 \Rightarrow y=0, z=0 \Rightarrow x^2/a^2 = 1 \Rightarrow x = \pm a \Rightarrow \text{Distance } d = a$

Case 2: $\lambda = -b^2 \Rightarrow x=0, z=0 \Rightarrow y = \pm b \Rightarrow \text{Distance } d = b$

Case 3: $\lambda = -c^2 \Rightarrow x=0, y=0 \Rightarrow z = \pm c \Rightarrow \text{Distance } d = c$

Maximum distance from origin is along the major semi-axis a ; minimum distance is along the minor semi-axis c .

Question 24 (Unit 4): Full Fourier Expansion of Discontinuous Square Wave**6 Marks**

Obtain the Fourier series expansion for the square wave function: $f(x) = -k$ for $-\pi < x < 0$, and $f(x) = +k$ for $0 < x < \pi$. Hence deduce: $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \pi / 4$.

Question Square Wave Symmetry: $f(x)$ is an odd function $\Rightarrow a_0 = 0$ and $a_n = 0$.

STEP-BY-STEP SOLUTION:**Solution Fourier Sine Coefficient Derivation:**

$$b_n = \frac{2}{\pi} \int_0^\pi k \sin(nx) \, dx = \frac{2k}{\pi} \left[(-\cos(nx) / n) \right]_0^\pi = \frac{2k}{n\pi} [1 - (-1)^n]$$

$$\text{For even } n: b_n = 0; \text{ For odd } n: b_n = \frac{4k}{n\pi}$$

$$f(x) = \frac{4k}{\pi} \left[\sin(x) + (\sin(3x) / 3) + (\sin(5x) / 5) + (\sin(7x) / 7) + \dots \right]$$

$$\text{Deduction: At point of continuity } x = \frac{\pi}{2}, \quad f\left(\frac{\pi}{2}\right) = k:$$

$$k = \frac{4k}{\pi} \left[\sin \frac{\pi}{2} + \left(\sin \frac{3\pi}{2} / 3\right) + \left(\sin \frac{5\pi}{2} / 5\right) - \dots \right] = \frac{4k}{\pi} \left[1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots \right]$$

$$\Rightarrow 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4} \quad (\text{Gregory-Leibniz Series Verified!})$$

The Fourier expansion contains only odd harmonic sine components, rigorously yielding the classical $\pi/4$ series identity.